

Evaluation of Safety Visualizations in Clinical Trial AE Reporting

4th March 2026

This survey explores how visualization styles affect interpretation of AE data. All figures use the same dataset; only the visual encoding varies. The goal is to assess how design choices influence experts' perception of safety signals.



* Erforderlich

Introduction & Consent

You are invited to participate in a short expert survey on the interpretation of adverse event safety data visualizations used in clinical trials.

Participation is voluntary and anonymous and no personally identifying information will be collected. The survey takes approximately 10-12 minutes to complete.

By continuing, you confirm that you are at least 18 years old and consent to participate in this study.

1

Do you consent to participate in this study? *

☐ Yes

☐ No

Background

2

Which best describes your professional background?

- ☐ Clinical research / clinical trials
- ☐ Biostatistics / data science
- ☐ Pharmacovigilance / safety
- ☐ Regulatory affairs
- ☐ Other

3

How many years of experience do you have interpreting clinical safety data?

- ☐ <2 years
- ☐ 2-5 years
- ☐ 5-10 years
- ☐ 10 years
- ☐ Prefer not to say

Visualization Blocks

Data Context

All figures are based on a comparison between Xanomeline High Dose and Placebo. Adverse events are shown at the Preferred Term (PT) level. Across all visualizations:

- Each subject contributed at most once per PT.
- Only PT with maximum incidence of $\geq 5\%$ in any treatment arm are included ("5% pure" threshold).
- The same set of PTs is used consistently across all figures.
- The number of subjects (N) per treatment arm remains constant across all figures.

Task Instructions

You will be shown a series of visualizations that present the same underlying adverse event data using different graphical methods (e.g. incidence comparisons, severity-based signal detection, temporal patterns, co-occurrence structure or multivariate associations).

For each visualization, please answer the accompanying questions based solely on what you see in the figure.

There are no right or wrong answers. We are interested in how clearly and effectively each visualization communicates safety-relevant information to you as an expert.

What to Focus On

When reviewing each figure, consider aspects such as:

- How easily potential safety signals can be identified
- How clearly differences between treatment and placebo are conveyed
- Whether the visualization supports rapid understanding or requires detailed inspection
- How confident you feel drawing conclusions from the display

You do not need to perform calculations or recall specific numeric values unless explicitly prompted.

Important Notes

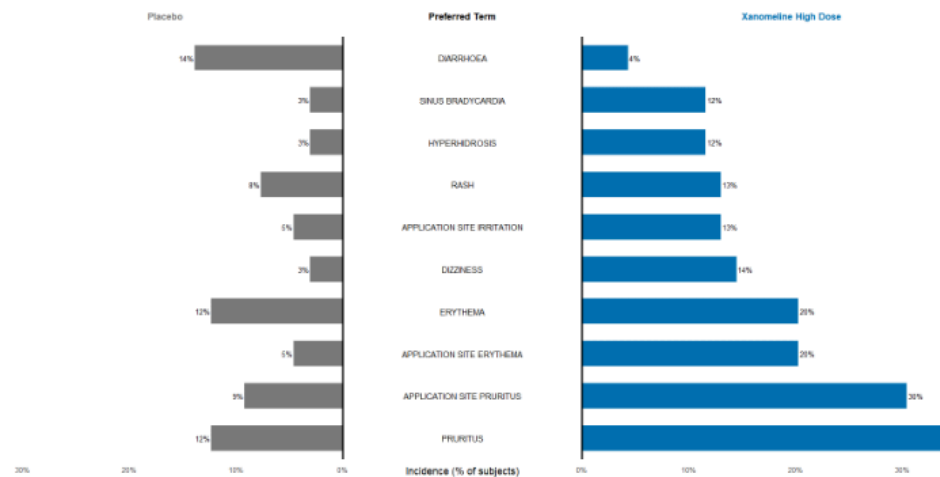
- The survey focuses on visual interpretation, not clinical decision-making.
- Figures are intentionally standardized to avoid differences caused by formatting or scaling.
- Some visualizations may highlight patterns (e.g. clustering or timing) rather than effect size of statistical significance.
- Please rely on your professional judgement when responding.

Butterfly Plot

Incidence of common adverse event preferred terms in Xanomeline High Dose versus Placebo

This figure shows a side-by-side comparison of adverse event incidence by Preferred Term (PT). Each row represents one PT, with Placebo shown on the left (grey bars) and Xanomeline High Dose on the right (blue bars). Bar length and the percentage labels indicate the proportion of subjects experiencing each event; the vertical black line in each treatment panel marks 0%. Only PTs with an incidence of at least 5% in one or both arms are shown, and PTs are ordered to highlight more frequent events and/or larger differences rather than alphabetically. When reviewing the figure, please focus on which PTs are most common in each arm, where the largest absolute differences between arms appear, and whether differences are limited to a small number of PTs or observed across multiple PTs.

PT Incidence (all grades) | 5% any-arm threshold
Placebo vs Xanomeline High Dose



4

Which Preferred Term shows the largest difference in incidence between treatment arms? *

- ☐ Application Site Pruritus
- ☐ Pruritus
- ☐ Rash
- ☐ Cannot determine from this plot
- ☐ Application Site Erythema
- ☐ Dizziness

5

For Pruritus, which statement best describes the comparison between High Dose and Placebo? *

- ☐ Higher incidence in Placebo
- ☐ Higher incidence in High Dose
- ☐ Similar incidence in both arms
- ☐ Cannot determine from this plot

6

How confident are you in your interpretation of this butterfly plot? *

- ☐ Very confident
- ☐ Confident
- ☐ Moderately confident
- ☐ Slightly confident
- ☐ Not confident at all

7

How much would you trust this visualization for exploratory safety analysis? *

- ☐ Completely
- ☐ Mostly
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

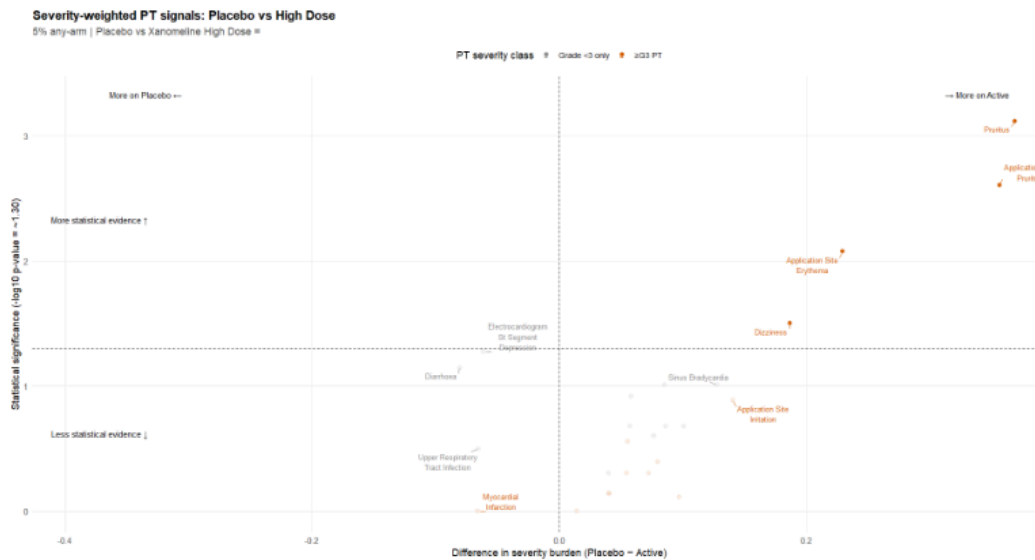
How easy was it to interpret this visualization? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Volcano Plot

Severity-weighted adverse event signals combining incidence, grade and statistical evidence

This figure compares adverse events between Placebo and High Dose by combining how often an event occurs and how severe it is into a single measure. Each point represents one adverse event (Preferred Term). The horizontal position shows the difference in severity burden between treatments: points to the right indicate events that are more pronounced under High Dose, while points near the center indicate little or no difference between treatments. The vertical position reflects the statistical strength of the difference, with higher points indicating stronger evidence. The dashed horizontal line marks a commonly used reference threshold for statistical evidence. The figure is intended to support a structured comparison across events and to help identify whether observed differences appear focused on specific types of events or spread more broadly.



9

Which adverse event represents the strongest overall safety signal in this plot? (Consider both severity burden and statistical strength) *

- ☐ Pruritus
- ☐ Application Site Erythema
- ☐ Rash
- ☐ Cannot determine from this plot
- ☐ Dizziness
- ☐ Application Site Pruritus

10

Which of the following statements is correct according to the plot? *

- ☐ Most statistically significant signals are grade <3 only
- ☐ Most statistically significant signals are grade ≥ 3 events
- ☐ Grade ≥ 3 and grade <3 events are evenly distributed
- ☐ Cannot determine from this plot

11

How confident are you in your interpretation of this volcano plot? *

- ☐ Very confident
- ☐ Confident
- ☐ Moderately confident
- ☐ Slightly confident
- ☐ Not confident at all

12

How much would you trust this visualization for exploratory safety analysis? *

- ☐ Completely
- ☐ Mostly
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

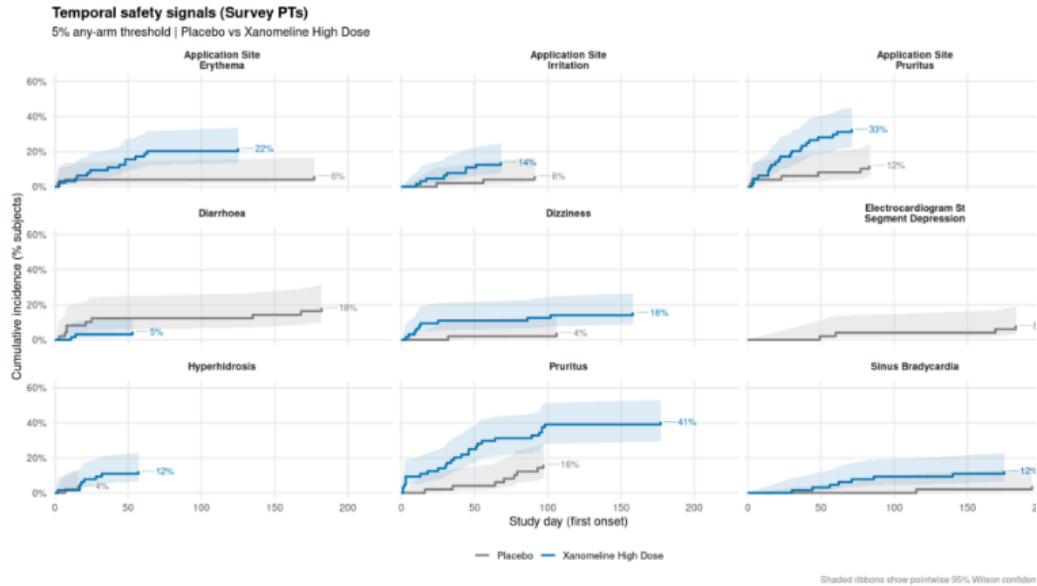
How easy was it to interpret this visualization? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Temporal Plot

Cumulative incidence of first adverse event onset over study time for selected preferred terms

Please look at each panel (Preferred Term). The two step lines show, over study time, the percentage of subjects who have experienced their first event of that term. When a line steps up, additional subjects reported their first onset around that day. The number at the end of each line is the final cumulative percentage by the end of follow-up. Compare the two lines to judge (1) which arm reaches a higher final percentage, and (2) whether events tend to occur earlier (steeper rise early) or later (gradual rise) in one arm versus the other.



14

Which adverse event shows the earliest separation between High Dose and Placebo? *

- ☐ Application Site Erythema
- ☐ Cannot determine from this plot
- ☐ Pruritus
- ☐ Diarrhoea
- ☐ Dizziness
- ☐ Application Site Pruritus

15

Based on the visualization, which statement best describes the timing of adverse event onset? *

- ☐ Adverse events tend to occur earlier in Placebo
- ☐ Adverse events tend to occur earlier in the active treatment arm
- ☐ Timing appears similar between arms
- ☐ Cannot determine from this plot

16

How confident are you in your interpretation of this temporal plot? *

- ☐ Very confident
- ☐ Confident
- ☐ Moderately confident
- ☐ Slightly confident
- ☐ Not confident at all

17

How much would you trust this visualization for exploratory safety analysis? *

- ☐ Completely
- ☐ Mostly
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

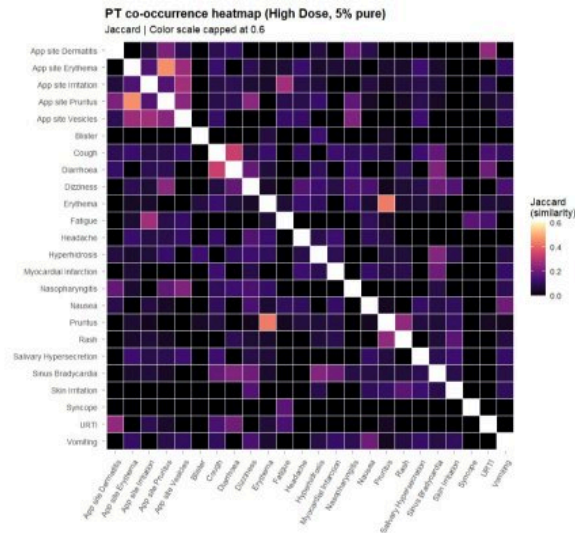
How easy was it to interpret this visualization? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Jaccard Similarity Heatmap

Frequency-adjusted similarity of adverse event co-occurrence patterns

Each square shows how often two Preferred Terms occur together in the same subjects. Dark colors mean the two terms rarely appear together, while brighter colors mean they frequently co-occur in the same individuals. The scale reflects similarity, not frequency or severity. Use this view to identify groups of terms that tend to cluster within subjects, rather than isolated events.



19

Which pair of adverse events shows the strongest tendency to co-occur in the same subjects? *

- ☐ Cough + Diarrhoea
- ☐ Cannot determine from this plot
- ☐ Dizziness + Headache
- ☐ Application Site Erythema + Application Site Pruritus
- ☐ Pruritus + Rash

20

Which statement best summarizes the co-occurrence pattern shown? *

- ☐ Systemic and local events strongly cluster together
- ☐ The plot does not show meaningful structure
- ☐ Adverse events occur largely independently
- ☐ Dermatologic adverse events tend to cluster together

21

How confident are you in your interpretation of this heatmap? *

- ☐ Very confident
- ☐ Confident
- ☐ Moderately confident
- ☐ Slightly confident
- ☐ Not confident at all

22

How much would you trust this visualization for exploratory safety analysis? *

- ☐ Completely
- ☐ Mostly
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

How easy was it to interpret this visualization? *

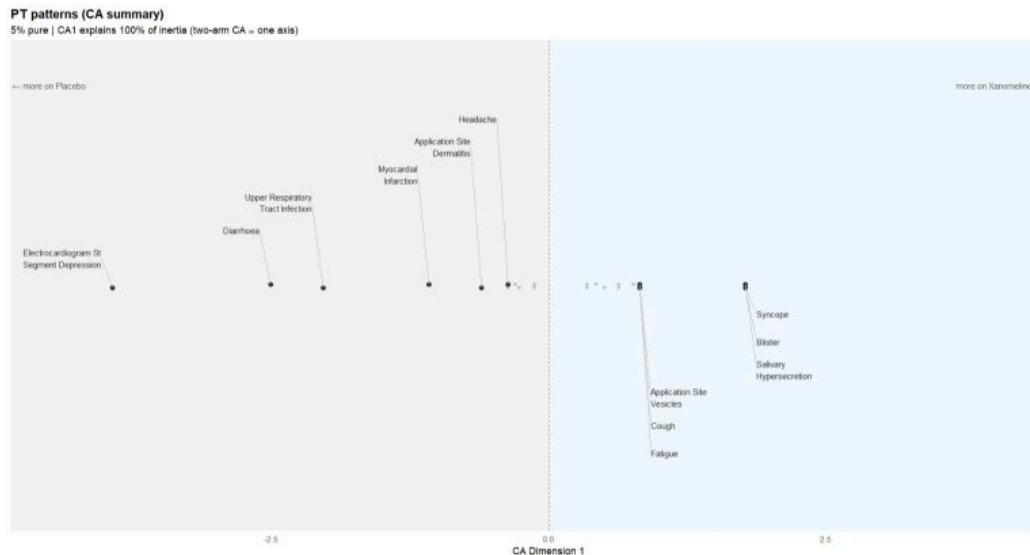
- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Correspondence Analysis (CA) Biplot

Correspondence analysis showing relative association of adverse event profiles with treatment arms

This chart summarizes which Preferred Terms (PTs) are more closely associated with Placebo versus Xanomeline High Dose based on overall PT patterns across the two arms. Each point represents one PT. Points to the left of the dashed zero line are relatively more associated with Placebo, while points to the right are more associated with High Dose. PTs near the center show similar patterns across both arms. The horizontal position shows how strongly a PT helps distinguish between the two arms. In this two-arm analysis, the single axis shown captures all meaningful differences between arms (100% of the inertia), meaning no additional dimensions are needed to describe the arm-level contrast.

Only the most strongly separating PTs are labeled to keep the figure readable.



24

Which preferred term appears most strongly associated with Xanomeline High Dose? *

- ☐ Myocardial infarction
- ☐ Upper Respiratory Tract Infection
- ☐ Diarrhoea
- ☐ Application Site Vesicles
- ☐ Cannot determine from this plot

25

Which of the following preferred terms appears more associated with Placebo than with Xanomeline High Dose? *

- ☐ Diarrhoea
- ☐ Application Site Vesicles
- ☐ Syncope
- ☐ Fatigue
- ☐ Cannot determine from this plot

26

How confident are you in your interpretation of this heatmap? *

- ☐ Very confident
- ☐ Confident
- ☐ Moderately confident
- ☐ Slightly confident
- ☐ Not confident at all

27

How much would you trust this visualization for exploratory safety analysis? *

- ☐ Completely
- ☐ Mostly
- ☐ Moderately
- ☐ Slightly
- ☐ Not at all

How easy was it to interpret this visualization? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

Final Comparison Questions

29

Which visualization was easiest to interpret overall?

	Very easy	Easy	Neutral	Difficult	Very difficult
Butterfly Plot	☐	☐	☐	☐	☐
Volcano Plot	☐	☐	☐	☐	☐
Temporal Plot	☐	☐	☐	☐	☐
Jaccard Heatmap	☐	☐	☐	☐	☐
CA Biplot	☐	☐	☐	☐	☐

30

Which visualization(s) would you most trust as an alternative to standard AE tables? *

- ☐ Butterfly Plot
- ☐ Volcano Plot
- ☐ Temporal Plot
- ☐ Jaccard Heatmap
- ☐ CA Biplot
- ☐ None of the above

Optional Free Text

31

If you have any comments of the visualizations or suggestions for improvement, please enter them here.

Dieser Inhalt wurde von Microsoft weder erstellt noch gebilligt. Die von Ihnen übermittelten Daten werden an den Formulareigentümer gesendet.

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